Why it turns on the spot, stops at the edge and sees the room

The working behind every number on the sheet approved 2026-11-09: how a tank turns on brass, what the gearbox trades, four switches a tread, one cell and its charge, why the drums are two thick halves, a servo that knows where it is, light timed to the picosecond, an eye of 64 rangers that the head sweeps, a light that does not fade, a blink and a face, why it does not tip, and the frame.
Read after Mars Rover Build. Nothing measured yet: every figure is derived from datasheet and product-page numbers and the sheet's dimensions, and each assumption is a named number in the generator.

The whole rover in one breath

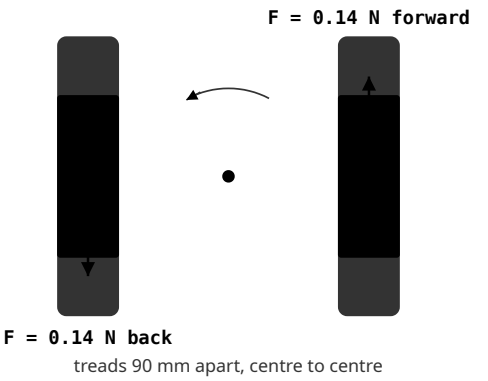
Two brass treads on four aluminium drums skid it around the desk. Two 150:1 gearmotors turn them at 56 rpm from one 3.7 V cell. The head, the camera board with an 8×8 laser ranger beside it, turns on a servo standing in the frame's collar; a laser on the left front post times its own light off the desk; a regulator holds the flashlight on the right post at 350 mA. One 950 mAh cell runs all of it for about 1.9 hours of exploring.

Every choice on the approved sheet follows from a piece of physics. This page takes them in the order the sheet made them: how a tank turns, what the gearbox trades, how the controller drives the treads, one cell and its charge, why the drums are two thick halves, the servo, the laser, the 8×8 eye that the head sweeps, the flashlight, the beacon and the face, why it does not tip, and the frame.

How a tank turns without steering

A tank has no steering. To turn, one tread runs forward and the other back, and both contact patches scrub sideways across the desk. That scrubbing is the whole cost of a turn.

Friction resists it. Each tread's 52 mm run carries half the rover's 3.22 N; sliding friction is the same everywhere on it, but its lever about the turning centre grows with the distance. Add it up and the resisting moment is μWL/4: 12.6 N·mm with brass on wood, μ about 0.3.



The rover pivoting left: the right tread drives forward, the left back, and the couple between them, 0.14 N a tread across 90 mm, beats the friction of the scrubbing runs.

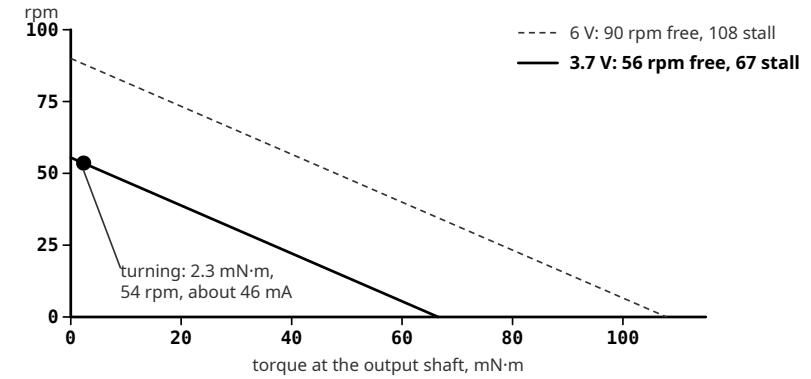
Each tread pushes 0.14 N to do it, 2.3 mN·m at its drum. The motor has 67 mN·m at stall: a 29-fold margin. That is why 10-gram motors turn a 328 g rover on the spot, and why the heavier wide drums cost nothing here.

Brass matters both ways. Silicone, μ about 1, would need 41.9 N·mm, more than three times as much, but it grips: it climbs slopes up to 45°. Brass slips on a nudge and climbs nothing steeper than 17°, a desk toy's trade. Pivoting, the treads' outer corners sweep 73 mm from the centre: give it a 16 cm circle of clear desk.

What the gearbox trades

A small DC motor spins fast and pushes weakly. Its torque falls in a straight line from stall to nothing at its free speed, and both ends scale with the voltage. At 3.7 V this one gives 56 rpm free and 67 mN·m at stall, 62 % of its 6 V figures.

The gears multiply torque by 150 and divide speed by 150. The tread moves with its pins, on a 33.4 mm circle round each drum: 97 mm/s free and about 83 on the desk. Slow enough to drive by camera, quick enough to cross a desk in ten seconds.



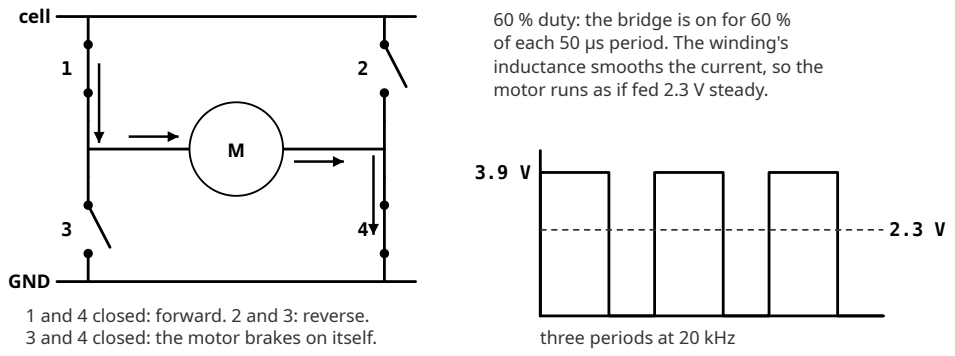
The motor's line at 6 V (dashed) and at the cell's 3.7 V. Turning on the spot uses 4 % of the stall torque, so the motor barely slows.

Current is the other half of the choice. Current follows torque, so a stalled motor draws its stall current: 222 mA at 3.7 V for this low-power winding. Two stalled together draw 444 mA, far from the cell's 2 A cut-out. The high-power winding of the same size stalls near 1 A at this voltage; two of those plus the camera would trip the cell the first time a tread jammed.

Four switches a tread, told over two wires

An H-bridge is four switches around the motor. Close one diagonal pair and current flows forward; close the other and it reverses; close both bottom switches and the motor brakes on itself. The motor controller holds one bridge per tread.

Speed is the share of time the bridge is on. It switches about 20 000 times a second; the winding's inductance smooths the current, so the motor sees the average: 60 % duty from a 3.9 V cell runs it like 2.3 V steady. Twenty kilohertz is above hearing, so the motors do not whine.



Switches 1 and 4 closed, the current's path in orange. Right: the bridge's output at 60 % duty, and the average the motor responds to.

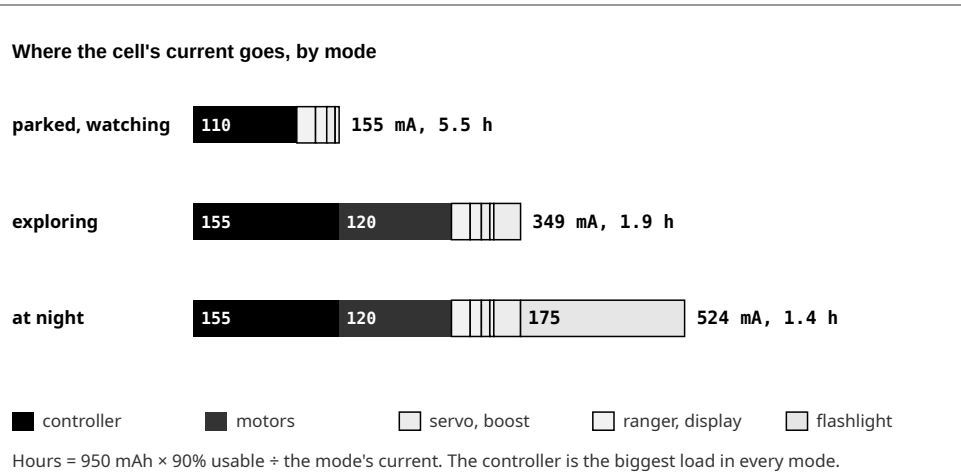
Its gate drivers work from 1.8 V, which is why this controller and not its 4.5 V twin: that one would not start on a single cell. It is also a small computer. It ramps the speed, reads the cell's voltage, and stops both bridges if the head goes quiet for 200 ms. At full speed that is 17 mm of travel before a crashed head stops the treads.

It listens on I²C: two lines that any device can pull low and resistors pull high. The ranger's, the display's and the controller's pull-ups in parallel make about 2.4 kΩ. With roughly 100 pF of wire and pins, a line rises in about 205 ns, inside the 300 ns that 400 kHz allows. Two shared wires carry everything on the frame, which is why the neck is seven wires and not twenty.

One cell for everything

The cell holds 3.4 Wh: 950 mAh at 3.6 V on average, from 4.2 V full to about 3.0 V empty. Everything runs straight from it: the motors through their bridges, the head's 3.3 V regulator, the servo's boost, the flashlight's regulator.

Exploring, the controller with its camera and Wi-Fi takes 155 mA, about a third; the 8×8 ranger 100; the two motors 120; the ranger, the display and the servo the rest. That is 449 mA and 1.9 hours. The flashlight at its cap adds 175 mA and brings it to 1.4.



The three modes as stacked bars, in milliamps from the cell.

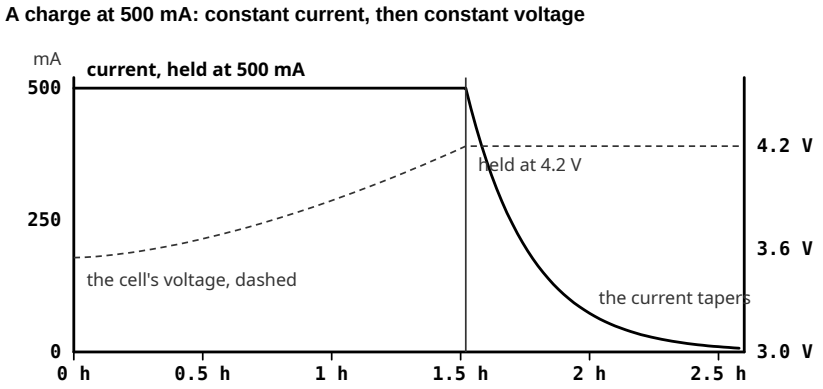
A cell sags under load. About 0.15 Ω sits between its chemistry and the wires: the cell itself, its protection and the holder's springs. A 1 A burst costs 0.15 V. The head's regulator needs about 0.2 V of headroom, so the firmware

parks the rover at 3.4 V rather than 3.0: there, at the camera's peak, the controller still gets about 3.1 V; much lower and it would brown out before the cell is empty.

The 220 µF beside the boost is a local store. The flashlight pulls 350 mA in pulses; over one 50 µs pulse the capacitor gives the charge and droops only 80 mV, so the cell's wires carry the average, not the pulses. Up the neck, the 26 AWG pair, 0.13 Ω a metre and about 7 cm long, loses 9 mV at the head's 0.52 A peak.

Charging: constant current, then constant voltage

A lithium cell charges in two steps. First a constant current, here 500 mA, a little over half the cell's capacity per hour; the voltage climbs to 4.2 V in about 1.5 h. Then the charger holds 4.2 V and the current tapers as the cell fills, and it stops when the current has fallen to a small fraction. About 2.2 h in all: red light, then green.



Current (solid) and the cell's voltage (dashed) through a full charge.

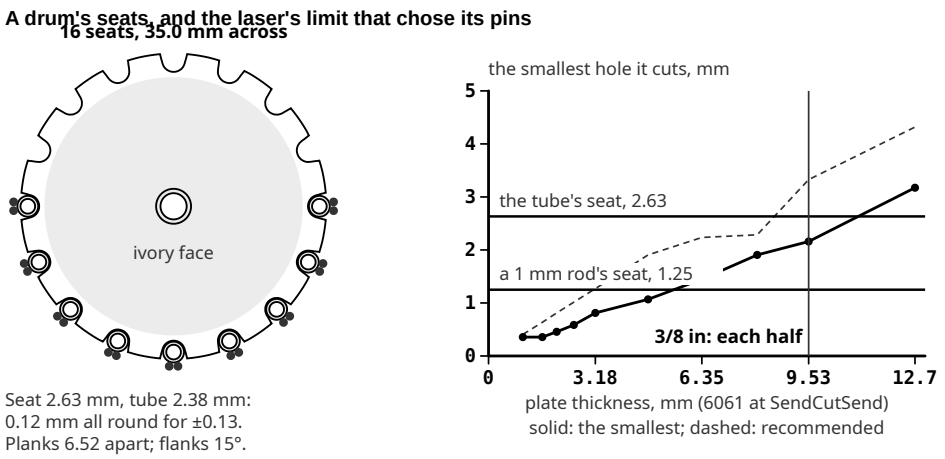
Why the switch must be off. This charger has no power path: whatever is connected to the cell is connected to the charger. With the rover on, the controller and the ranger take about 150 mA of the 500, the current never tapers to the stopping point, and the charger holds the cell at 4.2 V for as long as the cable is in. Switch off, and the charger sees the cell alone.

Why not the controller's own charger: it gives 100 mA, which is ten hours, and only while the rover sleeps. The 500 mA board turns (5 − 3.8) V × 0.5 A = 0.6 W into heat in a chip the size of a grain of rice. It throttles itself if it gets hot, and it stands in free air in front of the cell so it rarely does.

Why the drums are two thick halves

The reference's drums are machined aluminium, about three-quarters as wide as they are across, with narrow grooves that each hold a rod. Ours come off a laser. A fibre laser cuts by melting a line through the plate and blowing the melt out of the bottom with a jet of gas. In thick plate the slot is deep and narrow: the beam spreads past its focus and the gas loses its push, so a notch much narrower than the plate is thick fills with re-frozen metal instead of opening. Every cutting service sets a smallest hole that grows with thickness; SendCutSend's for 6061 is 0.36 mm in 1.6 mm plate and 2.16 mm in 9.53 mm.

A seat for a 1 mm rod needs 1.25 mm, the rod plus the laser's ±0.13 mm. That cuts in plate up to 4.75 mm and no thicker. So the pin became a 3/32 in brass tube, 2.38 mm, in a 2.63 mm seat: inside the limit for 3/8 in plate. One plate can't go thicker either: at 1/2 in the smallest hole is 3.17 mm and the smallest bridge 4.83, room for only 11 coarse seats. Two 3/8 in halves glued face to face give 19 mm and keep 16 seats.



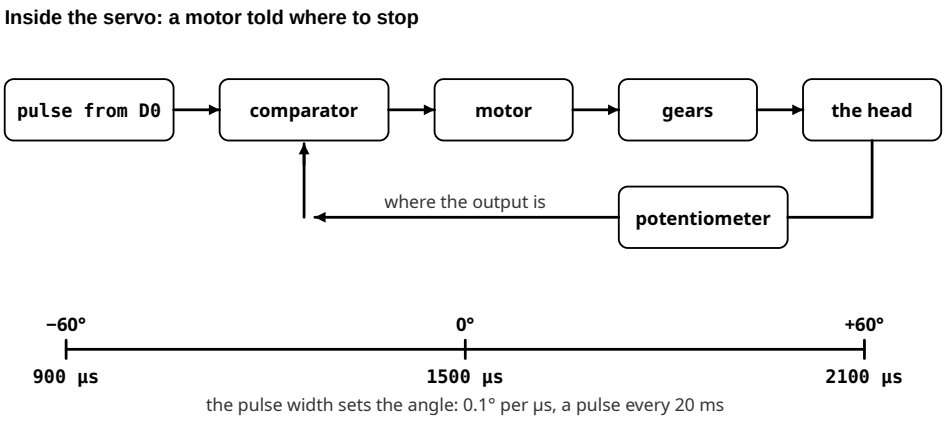
Left: the drum's cutting outline with the planks seated round its lower half. Right: SendCutSend's smallest hole against plate thickness, and the two seats; a seat can be cut where its line sits above the curve.

A tread round 16 seats is a 16-sided polygon, not a circle. As each pin rolls over the top of its seat the tread's speed rises and falls by 1 − cos(180°/16) = 1.9 %: the chordal action every chain drive has. On the desk the seats pass at about 13 a second, part of the brass's soft patter. Twenty seats would make it 1.2 %, eleven 4.1 %.

The seats' 15° flanks hold the pins in. A tooth pushing a pin also lifts it by tan 15° = 27 % of the push; the tread's weight and its neighbours hold it down. In a turn on the spot the push is only 0.14 N, spread over the pins in their seats: nothing an aluminium tooth notices.

A motor that knows where it is

A hobby servo is a small DC motor, a gear train, a potentiometer on the output shaft, and a comparator. Every 20 ms it gets a pulse; the width sets the target: 900 µs one end, 2100 the other, 1500 the middle. The comparator drives the motor until the potentiometer agrees.



The servo's loop, and the pulse widths the firmware sends: 0.1° for every microsecond.

The servo stands shaft up in the frame's collar, its body glued in, and the head's brass cradle is glued to its horn. When the servo drives its shaft, the head turns with it, 120° across the front of the rover.

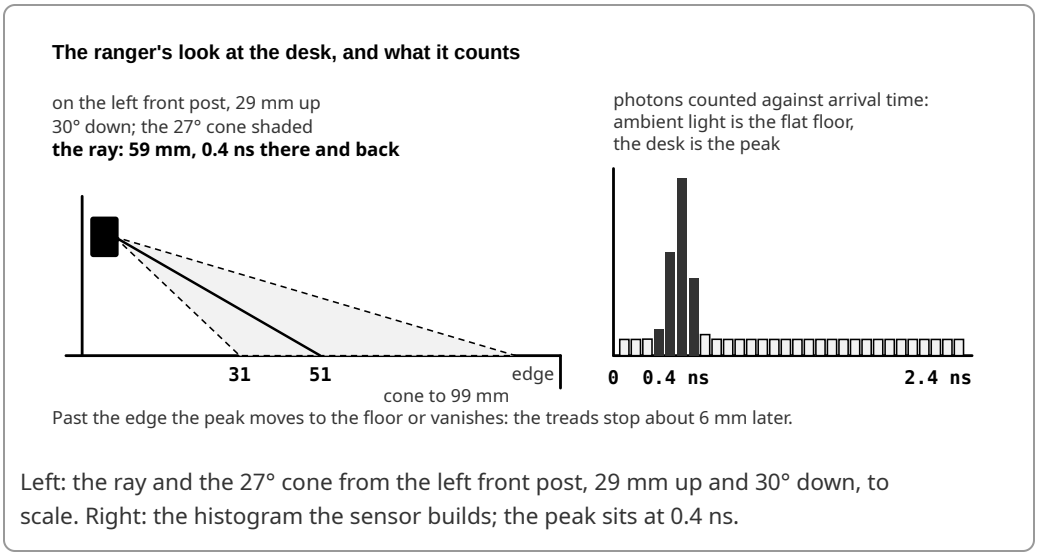
The head is easy for it. Its 9.1 g, a 35 × 15 mm block turning about its middle, swings through 60° in the rated 0.1 s on about 0.0048 kg·cm, 1.0 % of the servo's stall torque. The whole 120° takes about 0.2 s.

Why 5 V. The servo is rated 4.8 to 6 V; at 3.7 V its motor is weak and its electronics are out of their range. The boost converter stores energy in a small inductor and releases it at a higher voltage, about 85 % efficiently. The servo's 50 mA at 5 V costs about 79 mA from the cell; its 0.4 A stall about 636, inside the converter's 1.2 A switch.

Timing light off the desk

Light covers 30 cm in a nanosecond. The ranger fires invisible 940 nm laser pulses and times the photons that come back: the desk, 59 mm away along the ray, returns them in 0.4 ns. A millimetre of distance is 6.7 picoseconds.

No ordinary photodiode resolves that. The sensor uses single-photon avalanche diodes: tiny junctions biased past breakdown, where one arriving photon sets off an avalanche that can be timed. Over thousands of pulses it builds a histogram of arrival times, and the peak is the distance. Ambient light lands at random times and makes only a flat floor, which is why the reading holds in a bright room.



Left: the ray and the 27° cone from the left front post, 29 mm up and 30° down, to scale. Right: the histogram the sensor builds; the peak sits at 0.4 ns.

Over a flat desk the reading sits near 59 mm. At an edge it jumps to the floor, a metre down, or to no target at all, and that jump is the stop. At full speed and a reading every 20 ms the rover moves 1.7 mm between readings and about 6 mm before the treads are still. The warning, 5.1 cm, is 9 times the stop.

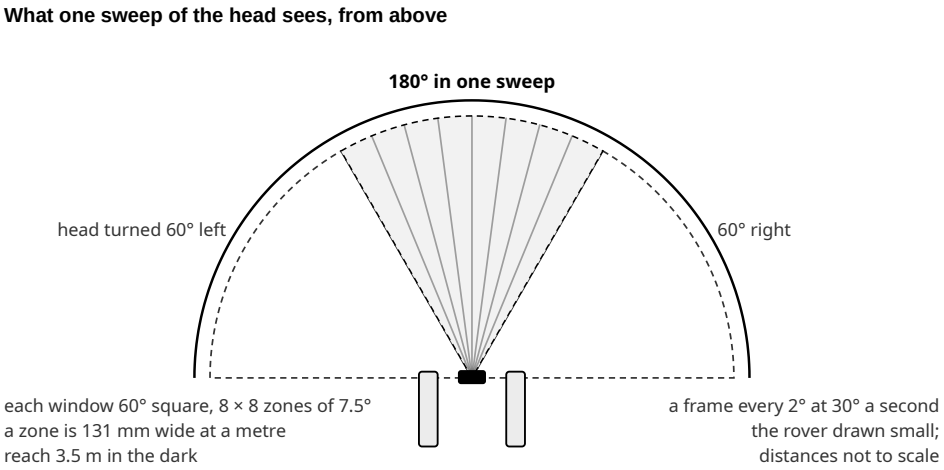
The cone's near edge meets the desk 31 mm out, 43 mm away along its ray: right at the sensor's 40 mm minimum, so the full cone's nearest returns are its least sure. Narrowing the 16 × 16 receiving array to a smaller region, about 15°, moves the near edge to 38 mm out and 48 mm away, clear of it, and tightens the watch to 38 to 71 mm. The ranger is fixed to the frame and does not turn with the head, so it always watches the way the treads are going.

An eye of 64 rangers that the head sweeps

The ranger on the post answers one question: how far is the desk. The 8×8 ranger in the head answers sixty-four at once. It is the same physics, a 940 nm laser timed by single-photon diodes, with a lens in front of the receiving array that maps a 60° square of the room onto 8 × 8 zones. Each zone sees a 7.5° slice, 131 mm wide at a metre. Its farthest wall, 3.5 m away in the dark, returns light in 23.3 ns. Daylight's infrared shortens that reach: the flat floor of stray photons rises until a faint return no longer stands above it.

One look covers a 60° window. The head turns 120°, so a sweep from one side to the other covers 180°, everything in front of the rover. Turning at 30° a second, a sweep takes 4 s and 60 frames at 15 a second, one every 2°: finer

than a zone, so the frames overlap and nothing between them is missed. Each frame goes to the desktop with the head's angle, which puts its 64 distances in their places on a map of the room.



From above: the head's 60° window at both ends of its turn and in the middle, the eight zones drawn in the middle one, and the 180° they cover between them.

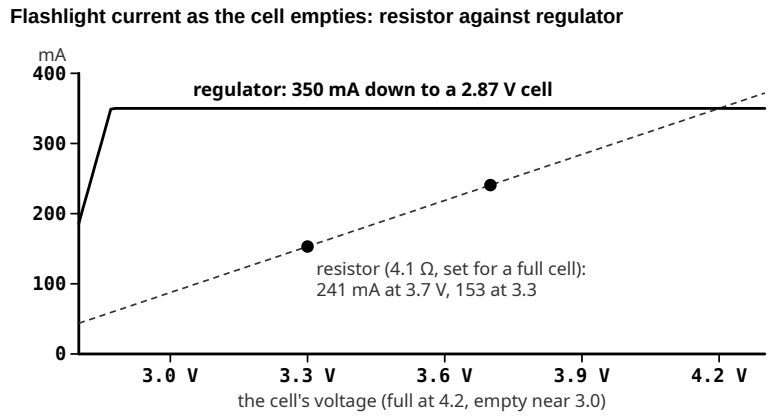
Why it has its own two wires. Every ST ranger of this kind answers at the same I²C address, 0x29, when it powers up, and the laser ranger on the post is one of them. Two devices at one address on one bus would answer together. The usual cure holds one in reset at boot and gives the other a new address, which costs a wire down the neck; a second bus in the head costs only two of the controller's spare pins, so the neck stays at seven wires.

Why it takes 3.3 V on its VDD pin. The board's level shifters pull its I²C lines up to whatever feeds VIN. Fed from the cell, they would sit at up to 4.2 V on the controller's pins, whose absolute maximum is 3.6 V. Fed on VDD from the controller's own 3.3 V, the lines sit at 3.3 V and the board's regulator is bypassed. It draws about 100 mA while it ranges, which the head's linear regulator takes straight from the cell; at worst the 3.3 V rail carries 572 mA, 82 % of its 700.

A light that does not fade with the cell

An LED's current climbs steeply with its voltage, so a resistor makes the light follow the cell. A resistor sized for 350 mA from a full 4.2 V cell gives 241 mA at 3.7 V and 153 at 3.3: less than half the light by the end of a run.

The regulator is a current sink instead. It holds 350 mA through the LED whatever the cell does, as long as the cell stays 0.12 V above the LED's 2.75 V. From full to nearly empty, the beam does not change.



The same LED from a resistor (dashed) and from the regulator (solid), against the cell's voltage. The regulator only lets go below a 2.87 V cell, where the rover has already parked.

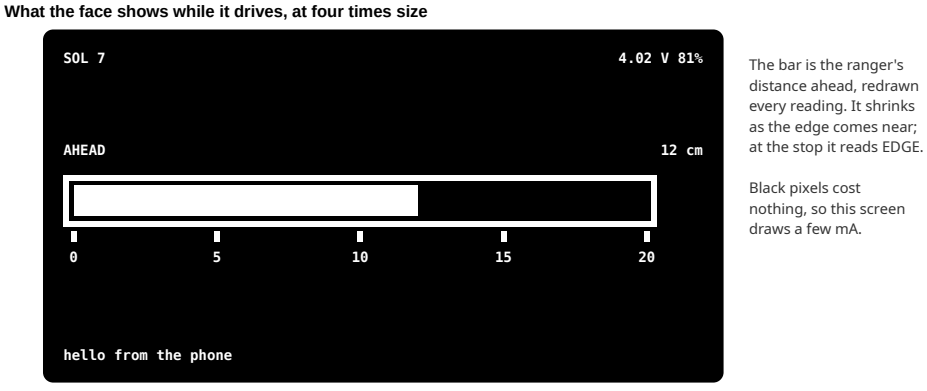
What it costs. At a full cell the regulator burns (4.2 − 2.75) V × 0.35 A = 0.51 W; the LED turns about a third of its 0.96 W into light and the rest into heat. At the 50 % cap the star carries 0.31 W of heat and warms about 19 °C. The drone runs the same light flat out in its propellers' wind; on the post it has no wind, so the cap stays.

The 10 mm lens folds the light into a 23° spot: about 937 candela on the axis, 3,747 lux on a spot 20 cm across at half a metre, several times a desk lamp. It points straight ahead from the right post; the camera sees by it when the head looks forward. The lamp's PWM runs at a few kilohertz, far faster than the camera exposes each row, so the picture shows no bands.

The beacon and the face

The eye adds up light for about a tenth of a second. A flash that long looks as bright as a steady light; a longer one looks no brighter, only longer. So the beacon's 100 ms every 2 s looks like its full 4 mA and averages 0.2 mA. The satellite's page has the whole of Bloch's law.

An OLED pixel is a tiny organic LED: it makes its own light, so a black pixel costs nothing. The rover's face is mostly black: a bar for the distance ahead, the cell, a line of text from the phone. The budget's 12 mA is a half-lit screen; this face lights about a tenth of its pixels.

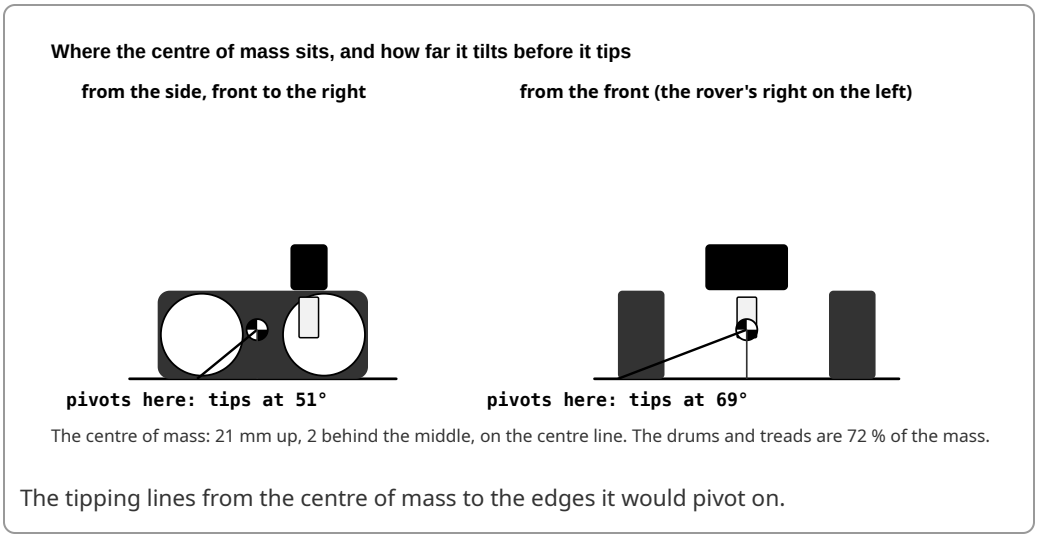


The 128 × 64 screen while driving, at four times size: the cell along the top, the ranger's distance ahead as a bar, a message from the phone.

Why it does not tip

The head is only 3 % of the mass, high up; the aluminium drums and the brass treads are 72 % of it, at axle height and below. The centre of mass comes out 21 mm up, 2 mm behind the middle, on the centre line.

A rover tips when its centre of mass passes over the edge of what it stands on. Sideways that edge is the outside of a tread, 69° of tilt away; backwards it is the rear of the contact run, 51°. A desk is flat; a 15 mm paperback under one tread tilts it 10°.

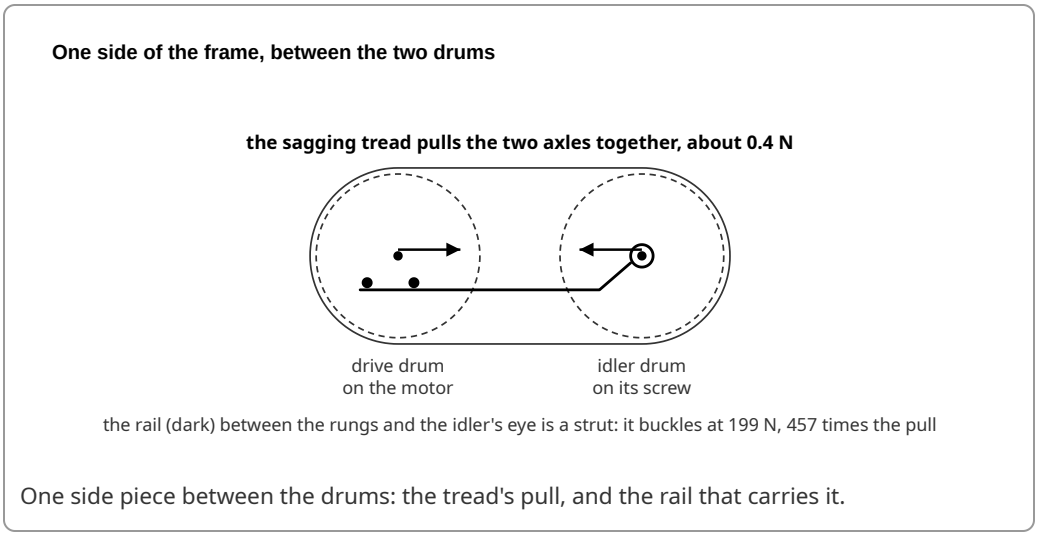


The tipping lines from the centre of mass to the edges it would pivot on.

The wide drums did this: they doubled the rover's weight but put it at the bottom, and the 90 mm between the treads makes it hard to roll. Braking from full speed in a tenth of a second is 0.08 g, the same as tilting 5°: nowhere near.

The frame: rods in compression

The brass tread does not stretch. Its last loop closes on the rover with about 1 mm of sag on the top run, and sag is tension: a chain weighing w per millimetre that sags δ over a span L pulls $T = wL^2/8\delta$. The tread weighs about 0.13 g a millimetre over the 52 mm span, so it pulls the two axles together with about 0.4 N.



One side piece between the drums: the tread's pull, and the rail that carries it.

The rail between them is a strut. A 1.5 mm stainless rod 49 mm long buckles at 199 N, 457 times the pull. Why 1.5 mm and not the 1 mm rod: stiffness goes with the fourth power of the diameter, so 1.5 mm is 5 times stiffer, and the idler's eye at the rail's end must not twist or the tread walks off. Why stainless: the family's frame material from the drone on, 193 GPa, hard to dent, and it solders with the acid flux on the bare frame before any part goes on.

What the frame does carry is the rover's weight. The front drums take 45 % of it, 0.7 N each, on M3 screws standing 13.5 mm out from their eyes. That bends each screw to about 7 MPa, against several hundred that stainless takes.

The numbers, and where they come from

NUMBER	WHAT IT IS
12.6 N·mm	the moment that resists a turn on the spot: μWL/4 = 0.3 × 3.22 N × 52 mm / 4
0.14 N	each tread's push to turn: the moment over 90 mm between the treads
29×	the motor's stall torque at 3.7 V (67 mN·m) over the turn's 2.3 mN·m
56 rpm	the gearmotor's free speed at 3.7 V: 90 rpm × 3.7 / 6
83 mm/s	on the desk: 56 rpm × π × 33.4 mm, less 15 % for load
222 mA	one motor stalled at 3.7 V: 0.36 A × 3.7 / 6
205 ns	an I^2C line's rise: 0.85 × 2.4 kΩ × 100 pF
17 mm	travel before the motor controller's 200 ms timeout stops the treads
3.4 Wh	the cell: 950 mAh × 3.6 V
80 mV	the 220 μF's droop over one 350 mA, 50 μs lamp pulse
2.2 h	a full charge: about 1.5 h at 500 mA, then the taper
2.16 mm	SendCutSend's smallest hole in 3/8 in 6061: why the pins are 2.38 mm tubes in 2.63 mm seats
1.9 %	the tread's speed ripple round 16 seats: 1 − cos(180°/16)
0.4 ns	the laser's round trip to the desk, 59 mm away
51 mm	where the ranger's centre ray meets the desk: 29 / tan 30°
6 mm	from the edge seen to the treads still: one reading plus half the ramp
7.5°	one zone of the 8×8 ranger: 60° / 8
180°	what one sweep of the head sees: its 120° turn plus the 60° window
572 mA	the 3.3 V rail at its worst, the 8×8 ranger included, of the regulator's 700
153 mA	what a resistor would give the flashlight at a 3.3 V cell, against the regulator's 350
19 °C	the star's rise at the 50 % cap: 0.31 W × 60 °C/W
937 cd	the beam: 139 lm × 0.85 into a 23° cone
69° / 51°	tipping, sideways / backwards, from a centre of mass 21 mm up
0.4 N	the tread's pull with 1 mm of sag over 52 mm: $wL^2/8\delta$
199 N	the rail's buckling load, 49 mm of 1.5 mm 304, against the tread's 0.4 N
7 MPa	an idler screw's bending stress, 0.7 N at 13.5 mm on an M3

Which part each idea lives in

- **skid steering, μWL/4, brass's low grip:** the 1.0 mm brass rod
- **the laser's limit, the seat, the chordal ripple:** the aluminium drum halves
- **the pin in its seat:** the 3/32 in brass tube
- **the torque line, 150:1, the stall current:** the micro metal gearmotor
- **the H-bridge, PWM, the 1.8 V gate drivers, the timeout, I²C:** the motor controller
- **3.4 Wh, the sag, the 3.4 V park:** the Nitecore 16340 cell
- **constant current then voltage, no power path:** the USB-C charger
- **the closed loop, the pulse widths:** the sub-micro servo
- **an inductor's stored energy, 85 %:** the 5 V boost regulator
- **time of flight, SPADs, the histogram, the cone:** the laser ranger
- **64 zones, the sweep, a second bus, VDD not VIN:** the 8×8 laser ranger
- **the current sink, the dropout, the heat:** the 350 mA LED regulator
- **candela, lux, the spot:** the cool-white LED star
- **Bloch's law:** the pre-wired LED chip
- **self-lit pixels, the distance bar:** the 0.96 inch display
- **the camera's rows, the Wi-Fi's share of the budget:** the controller with camera
- **Euler, the fourth power, sag tension:** the 1.5 mm stainless rod
- **the cantilevered axle:** the M3 × 25 screws

What comes next

Check out the carts: the DigiKey cart (pick USPS Ground Advantage), the Amazon page and the nickel-silver wire, the drum halves at SendCutSend. Then a bench test for each idea on this page before the frame is soldered: a drum on a motor with a strip of twelve planks, run at 3.7 V with its current read; the ranger held over a desk edge at the post's height and tilt; the 8×8 ranger on the head's second bus, turned by hand across a room; the servo on its boost turning the head; the flashlight through its regulator from 4.2 down to 3.0 V. Caliper the drums, the display, the holder and the head's boards, bend the frame on a 1:1 jig, and write the firmware.

Built by `docs/pages/gen/physics_page.py` from the dimensions, head, budget and drum in `build_page.py` (which imports `cad/drum.py`); it also writes `docs/PHYSICS.md`. Assumptions, each a number at the top of the script: μ 0.3 for brass on wood and 1.0 for silicone, a 52 mm contact run, a 33.4 mm pitch line, 0.15 Ω in the cell's path, 0.2 V of regulator headroom, 100 pF on the bus, 9.1 g of head turning as a 35 × 15 mm block, the LED as 2.6 V plus 0.43 Ω, 85% through the lens, 60 °C/W off the star, 1 mm of tread sag, a 30° a second scanning turn for the 8×8 ranger, 304 at 193 GPa, and SendCutSend's 6061 laser limits read 2026-11-07. The motor, servo, boost, ranger, charger and regulator figures are their makers' pages and datasheets. Nothing here has been measured on the rover itself.